

RESEARCH

Open Access



Multimodal sentiment analysis using image and text fusion for emotion detection

Uttam U. Deshpande^{1*}, Supriya Shanbhag¹, Amit Sukhasare¹, Mahendra M. Dixit², Rudragoud Patil³, Sangeeta Sangani^{4*}, Sowmyashree H. Srinivasaiah⁵, Swetha Goudar⁶ and Manjunath Managuli¹

*Correspondence:

Uttam U. Deshpande
uttamudeshpande@gmail.com
Sangeeta Sangani
sangeeta.sangani@manipal.edu

¹Department of Electronics and Communication Engineering, KLS Gogte Institute of Technology, Belagavi, Karnataka, India

²Department of Electronics and Communication Engineering, KLS VDIIT, Haliyaal, Karnataka, India

³Department of Computer Science and Engineering, KLS Gogte Institute of Technology, Belagavi, Karnataka, India

⁴School of Computer Engineering, Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, India

⁵Master of Computer Applications Department, Bangalore Institute of Technology, Bengaluru, India

⁶Department of MCA, KLS Gogte Institute of Technology, Belagavi, Karnataka, India

Abstract

Social media has become an essential platform for expressing personal experiences and emotions. Today's youth frequently share images that reflect their emotional states, including happiness, excitement, sadness, anxiety, and distress. Accurately analyzing these images using new frameworks can offer beneficial insights into the emotional well-being of individuals. Beyond mental health applications, image sentiment analysis has significant potential in marketing and advertising. Brands and Marketers can get a more comprehensive understanding of consumer sentiments and preferences by examining the emotional reactions elicited by visual content. For instance, companies can analyze images shared by customers to gauge sentiment towards their products and services. Positive or negative feedback expressed through images can offer practical insights for improving products and customer experience. Additionally, Sentiment analysis is one tool that marketers can use to gauge the effectiveness of their advertising campaigns. By analyzing the sentiments of images associated with a campaign, they can determine which aspects resonate most with the audience and adjust their strategies accordingly. Our research focuses on creating an advanced multimodal sentiment analysis system that combines BERT and Vision Transformers (ViT) to analyze textual and image data. High-precision sentiment classification is achieved by our technique using a preprocessed AllenTAN dataset from Hugging Face. It conducts sentiment analysis using BERT, creates captions for unlabeled photos, and uses OCR to retrieve embedded image text. The suggested ViT + BERT technique performs well with a variety of social network content. The proposed system achieves an accuracy of 96.91%, demonstrating its robust performance across diverse social media content and benchmark models. This technology has several uses, particularly in social media monitoring to promote mental health content, as teens frequently use visuals to describe their feelings.

Keywords Mental health, Image sentiment analysis, Emotion, AllenTAN, Vision transformers (ViT), Optical character recognition (OCR), BERT

1 Introduction

People's innate attitudes about a specific subject, person, or thing are known as their sentiments. It is useful for us to communicate, learn, and make decisions when we are aware of people's attitudes. With the popularity of social media and mobile devices,



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

individuals are posting text and images together to share their thoughts and feelings. Sentiment analysis uses text, audio, video, and picture analysis to gather people's viewpoints, opinions, attitudes, and emotions [1–3]. For instance, product-based sentiment analysis [4] is popular among consumers and e-commerce websites because it can assess users' emotional inclinations toward items and brands. Businesses may then utilize this information to enhance the quality of their products and brand image. A business or retailer may alter their operations accordingly based on the opinions of their customers regarding their product or brand. The government can benefit from social media sentiment analysis [5] in understanding public views or positions on important events or popular subjects. Public opinion can be influenced by government entities through the assessment of online voting. For this reason, researchers in Artificial Intelligence have been working to give machines the cognitive capacity to identify, understand, and communicate emotions over the past 20 years. Social media's rapid growth in popularity has had a profound impact on how people communicate and express their emotions in recent years, making websites like Facebook, Instagram, and Twitter vital channels for sharing experiences and sentiments via photos and captions. These platforms are now essential for how people express their emotions, particularly young people, who frequently do so by fusion of textual and visual aspects. The ability to evaluate and decipher these emotions could provide an invaluable understanding of the emotional health of the general population and open the door to prompt mental health interventions.

There are benefits and drawbacks to gathering and analyzing data from different sources when compared to single-modal sentiment analysis. Since text and visuals typically come in pairs in real life, information is frequently not unimodal. Multimodal sentiment analysis, which seeks to identify and analyze feelings from data with a variety of modalities, has garnered more and more attention from researchers in recent years [6]. Most early multimodal works rely on handcrafted features. Nevertheless, handcrafted features are usually created with a limited amount of human expertise and are not able to fully capture the high complexity of sentiment, which results in undesirable outcomes. Three categories can be used to organize various fusion techniques: early fusion, middle fusion, and late fusion. The primary means of achieving early fusion is feature extraction from multi-modal data, followed by splicing, weighting, and other fusion techniques. However, the fusion output may contain many redundant vectors, leading to information redundancy and dependence and a poor fusion effect. Convolutional neural networks have been used in multimodal sentiment analysis with positive results in recent years, thanks to the advancements in deep learning. The neural network is primarily used to achieve intermediate fusion, sharing the shared network's middle layer as it goes through the fusion process. After training each modality to select the one that best fits its needs, late fusion performs decision fusion using many classifiers to provide predictions. However, late fusion is not very good at coordinating correlations across the different modalities. However, most Convolutional Neural Networks (CNN)-based methods either fuse characteristics obtained from several modalities [7] or allow for an elementary understanding of the link between text and images [8]. Human emotions are mostly elicited by specific regions [9] in images and are conveyed through phrases in the commented text that goes along with the image. The current state of image sentiment analysis techniques still faces some unresolved issues that require immediate attention. For example, the accuracy of the image regions that feature extraction networks target is

insufficient, information redundancy arises from the straightforward feature extraction process for independent modalities, and the semantic correlation between multimodalities is not sufficiently analyzed and fused.

The motivation behind this research is driven by the increasing concerns about mental health, particularly among youth. Social media networks have observed a surge in posts reflecting emotions such as sadness, anxiety, and depression, underscoring the requirement for monitoring instruments and understanding emotional well-being through social media activity. Early detection of negative emotions can result in timely support and intervention, potentially treating mental health conditions before their escalation. This real-time sentiment analysis can function as a critical component in mental health monitoring systems, enabling experts in mental health to identify at-risk individuals and provide necessary support. This proposed work intends to promote content creation, improve marketing strategies, and improve mental health monitoring by providing a powerful tool for real-time sentiment analysis. We explore the possibility of applying cutting-edge machine learning methods to important real-world problems by integrating Vision Transformers (ViT) and Bidirectional Encoder Representations from Transformers (BERT) models in a single framework. The objectives of the proposed work are summarized as follows,

- i. Developing an integrated sentiment analysis system using ViT and BERT-based models. Based on visual content and associated language, this system will reliably classify photos into sentiment categories (positive, negative, and neutral) using deep learning algorithms.
- ii. Enhancing accuracy and efficiency of sentiment analysis through the integration of ViT for image feature extraction and BERT for textual sentiment classification. The goal is to achieve state-of-the-art performance in recognizing and interpreting emotional cues from images and associated captions.
- iii. Enable real-time sentiment analysis capabilities by providing instantaneous feedback on the emotional content of images, facilitating prompt responses and interventions where necessary.
- iv. Ensure the system is scalable to handle large volumes of images from different social media platforms. Integrate cross-modal analysis capabilities to correlate sentiment between images and accompanying text, enhancing the depth and accuracy of emotional understanding.
- v. Implement mechanisms for continuous learning and model adaptation to evolving trends and changes in user behavior on social media platforms. Extend sentiment analysis capabilities to support multiple languages, accommodating diverse global users and content across different linguistic contexts.

The structure of the paper is as follows: In Sect. 2, important past literature is reviewed. The proposed methodology is presented in Sect. 3. The system architecture and implementation techniques are described in Sect. 4. The experimental results and analysis are provided in Sect. 5. In Sect. 6, the research findings are summarized, and future work is suggested.

2 Literature overview

This section reviews sentiment analysis literature.

2.1 Multimodal sentiment analysis

In recent years, multi-modal sentiment analysis has drawn increasing attention, and it is an exceedingly challenging research area. Unlike single-modal sentiment analysis, multi-modal sentiment analysis incorporates several domains, including computer vision, natural language processing, and others. It is a field of study that involves multiple disciplines. The incorporation of both verbal and nonverbal information, such as visual and audio data, has boosted multimodal sentiment analysis to a prominent area of study. Historically, multimodal fusion and representation learning have been the focus of research.

Splicing, weighing, and other techniques are used to carry out the fusion process once the feature extraction from different modal information is accomplished by the early fusion method. Feature splicing creates a single bag of words [10] by merging text and visuals to create the final representation. The interactive information approach [11] gathers the underlying picture features and binary representation to extract text features. The resultant similarity-based neighbourhood classifier is then subjected to binary classification using the approach. Each modality's data is trained independently in late fusion, which then chooses the best classifier and provides the final fused result. First, distinct characteristics for text and images were retrieved using CNN [12]. Next, different emotions were predicted and examined using logistic regression. Finally, a weighted fusion technique with an average was used to conduct the final emotion prediction study. Modality-specific and modality-invariant representations were presented [13] for multimodal representation learning. Depending on the stage at which fusion is implemented, the information that is currently available on multimodal fusion can be divided into two categories: early fusion and late fusion. Early attempts at cross-modal fusion typically employed subtle attention mechanisms. A hybrid deep learning model [14] first performs fine-grained analysis on multi-modal data and then utilizes a decision-level multi-modal combination to categorize and output the data. A cross-modal converter [15] was proposed to learn cross-modal attention to improve the target modality. After learning the intra-modal representation, inter-modal fusion was accomplished using a tensor fusion network and post-fusion approaches. However, this method did not allow for any multimodal interaction. To solve this issue, a multi-tensor fusion method [15] based on cross-modal feature extraction was put forth. However, this method ignored any data transferred between the modalities and only roughly fused the image and text modalities during the feature fusion phase. The global context of modalities was overlooked despite the introduction of a multi-tensor fusion network [15] to depict interactions between many bimodal pairs. Additionally, to enhance model performance, a hybrid model that used a knowledge graph Long Short-Term Memory (LSTM) and CNN was developed [16]. LSTMs' temporal data memory functions and CNNs' local feature extraction powers are both utilized in this technique. However, the approach is dependent on the structure of the knowledge graph; therefore, processing complex linguistic phenomena such as puns and sarcasm may potentially generate instability in the model.

Creating a unified dataset was a critical component of our proposed research. We combined and preprocessed datasets from various sources to ensure consistency and accuracy in sentiment labels and image paths. This involved handling missing values, standardizing sentiment labels, and ensuring that image paths were correctly mapped to their corresponding sentiment labels. The comprehensive dataset served as the

foundation for training and evaluating models, ensuring that the system could accurately interpret a wide range of emotional expressions in images.

2.2 Vision transformers (ViT) and BERT-based models

When used to model sequential data, a Transformer is a non-looping sequence-to-sequence design that can provide better depth, efficiency, and outcomes than cyclic structures [17]. ViT and BERT are two models that are based on the Transformer. The ViT vision model, which is built on the Transformer architecture, performs exceptionally well [18, 19] in computer vision tasks with intriguing applications. It is predicated on the idea of breaking up an image into smaller plots and using a self-attention process to model and extract features from the images. The successful use of the Transformer for language models is BERT [20–22], which achieves exceptional accuracy in a range of text processing applications that extract features from text and picture data, respectively. The open-source pre-trained BERT and ViT models are employed to accomplish these tasks. A machine learning technique called multi-task learning (MTL) [23, 24] builds a model to learn several related tasks at once. It can somewhat increase the model's efficiency and ability for generalization. By considering the correlations and contrasts between different tasks, multitask learning can facilitate information transfer and interaction between them while building a suitable model architecture and loss function. For MSA, multi-task learning [25–27] has been widely utilized.

2.3 Enhancing multimodal sentiment analysis by integrating ViT-BERT

Significant progress has been made in multimodal sentiment analysis, especially with the use of deep learning and attention-based methods. Particularly in practical applications like social media analysis, e-commerce reviews, and content moderation, traditional unimodal methods frequently fall short of capturing the complete semantic and emotional complexity of information. A deeper comprehension of sentiment is provided by multimodal models, which integrate textual and visual data. The development of fusion techniques is highlighted in recent publications. A late fusion-based multi-task learning framework [26] demonstrated cutting-edge performance on the gathered multimodal dataset. A two-stage multi-task sentiment analysis framework [25] combined pre-trained models with multi-task learning strategies. This framework improved multimodal sentiment classification performance and produced positive findings on multimodal datasets. Multi-tensor fusion was presented [15] as a method to improve cross-modal representation learning and, consequently, the synergy between text and image inputs. A thorough analysis of current fusion techniques and benchmark datasets was presented [8], setting the stage for comparative analysis. A hybrid deep learning architecture tailored for sentiment-rich data was presented [16], which also showed gains in identifying subtle emotional indicators. A multi-task learning model that included specialized scoring and cascade [27] concurrently executed intra- and inter-modal interactions to improve multimodal sentiment representation skills. These models are developed using MSA, struggle to produce a comprehensive comprehension of the emotional content when the text fails to sufficiently convey the nuanced emotional aspects of the imagery.

The goal of our proposed work is to create a system for sentiment analysis of images by using ViT and BERT-based models to extract hidden text, create captions for images, and assess the overall sentiment. Applying transformer models to sequences of image

patches, vision transformers provide state-of-the-art performance and have become an excellent way to identify images. BERT-based models, on the other hand, have completely changed natural language processing by offering strong comprehension and production capabilities for human language. The suggested method seeks to offer a thorough comprehension of the emotions expressed through both textual and visual information by combining both technologies. This research intends to promote content production, improve marketing strategies, and improve mental health monitoring by providing a powerful tool for real-time sentiment analysis. The integration of ViT and BERT models in a unified framework demonstrates the potential of leveraging state-of-the-art machine learning technologies to address critical real-world challenges. Not only does this method offer sentiment categorization as positive, negative, or neutral, but it also delves deeper into specific emotions such as sadness, anxiety, anger, and depression for negative sentiments, and happiness, joy, and excitement for positive sentiments, thereby offering a nuanced understanding of user emotions. We test our model on the AllenTAN image sentiment dataset, which provides a carefully chosen collection of photos from various web sources and social media platforms.

3 Methodology

As shown in Fig. 1, the AllenTAN/image sentiment dataset [28] forms the core of our image sentiment analysis, offering a curated selection of images sourced from diverse social media platforms and online repositories. Each image in this dataset is meticulously annotated with sentiment labels, categorizing them into positive, negative, or neutral sentiments based on user interactions and annotations. This dataset is instrumental in training and evaluating our sentiment analysis model, providing a rich variety of visual content across different contexts.

The dataset possesses the following characteristics,

- i. *Source Diversity*: Images are collected from different platforms and online repositories, ensuring a broad representation of user-generated content.
- ii. *Sentiment Labels*: Each image is labeled with sentiment categories, enabling the model to learn and classify sentiments effectively.
- iii. *Metadata Enrichment*: Alongside sentiment labels, images are accompanied by metadata such as captions, tags, and timestamps. These metadata elements provide

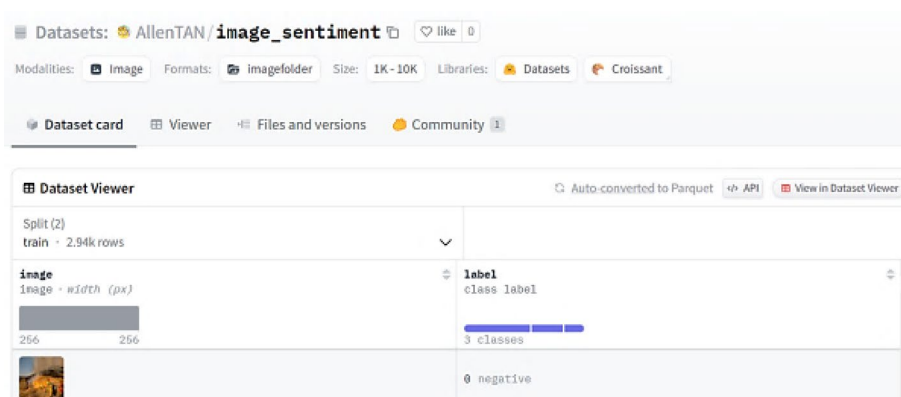


Fig. 1 AllenTAN dataset [28]

crucial contextual information, enriching the understanding of image content and sentiment nuances.

Sentiment labels are assigned through a manual annotation process where human annotators categorize images based on visual cues and accompanying textual information. Rigorous quality checks ensure the accuracy and consistency of sentiment annotations, minimizing labeling bias and errors. The AllenTAN/image sentiment dataset serves as the primary training set for our sentiment analysis model. Its diverse nature and comprehensive annotation facilitate robust model learning and generalization. This dataset's images are utilized to assess the models, measuring accuracy, precision, recall, and other evaluation metrics to assess sentiment classification effectiveness. Hugging Face provides the AllenTAN/image sentiment dataset, which has previously undergone pre-processing. This guarantees constant sentiment tagging, metadata integration, and image resolution. To improve model generalization, we supplemented this with an additional dataset of 10,000 images. This dataset augments the primary dataset by offering a broader variety of visual content across various domains and contexts. While not annotated with sentiment labels as comprehensively as the primary dataset, the supplementary dataset enriches our training corpus with additional diversity and variability.

3.1 Data preprocessing

A common format for raw picture data is created during image preprocessing so that deep learning models can use it to analyze sentiment. Images in the dataset often vary in resolution and size. Standardizing images to a common resolution, such as 224×224 pixels, ensures uniformity and compatibility across the dataset. This standardization simplifies the input requirements for neural networks, which typically expect fixed-size inputs for efficient processing. We perform normalization to adjust the pixel values of images to a standardized range, typically $[0, 1]$ or $[-1, 1]$. Normalizing pixel values enhances model stability during training and convergence, preventing the dominance of certain features due to differing scales. Data augmentation techniques are applied to increase the diversity and robustness of the training dataset. Augmentation methods such as rotation, flipping, cropping, and brightness adjustments generate variations of the original images, effectively increasing the dataset size and improving model generalization. The data preprocessing steps are described in Fig. 2.

3.1.1 Data cleaning

Data cleaning involves removing noise and irrelevant characters from textual metadata, ensuring that the processed text retains only the essential semantic content. Here are the common steps involved in text cleaning. HTML tags and special characters, such as '<', '>', and '&', are typically found in web-based textual data. These tags do not contribute to the semantic meaning and are removed to clean the text. Punctuation marks (e.g., ",", " ", "?") and numeric characters are often irrelevant for sentiment analysis tasks and can be safely removed. This step helps reduce the vocabulary size and computational complexity. Text normalization involves converting all characters to lowercase. This standardizes the text and ensures that words with different cases (e.g., 'Hello' and 'hello') are treated as identical tokens during tokenization.

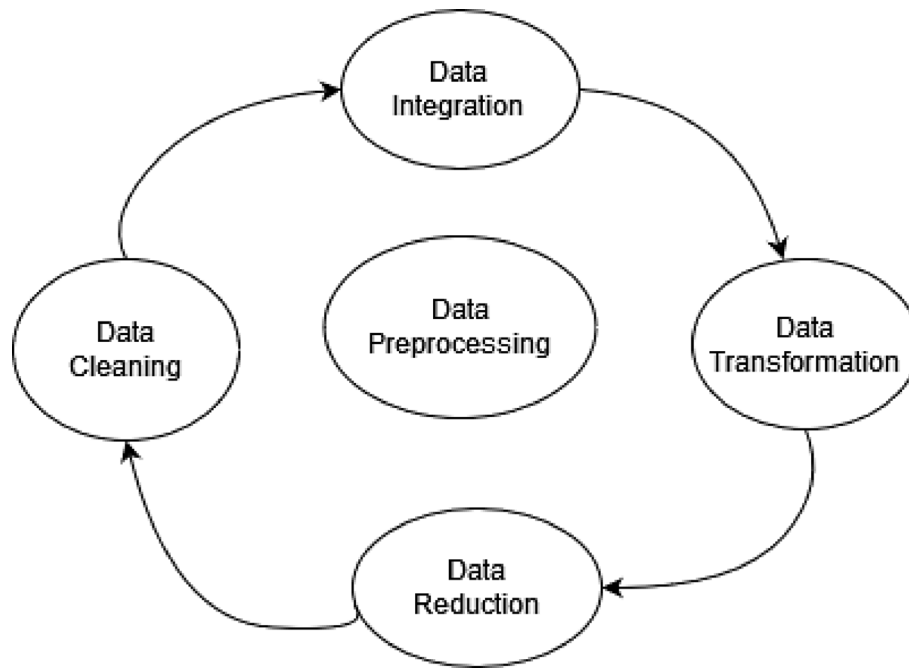


Fig. 2 Data preprocessing

3.1.2 Tokenization

Tokenization is the process of breaking down textual data into smaller units, such as words or sub-words, which will be converted into numerical representations. Effective tokenization is important for handling text data in deep learning models. Modern tokenization methods, particularly those used with transformer-based models, employ sub-word tokenization techniques that split words into meaningful subunits called tokens. This approach allows the model to handle rare words and out-of-vocabulary terms effectively. Transformer-based models like BERT shown in Fig. 3, use sub-word tokenization, such as WordPiece or Byte-Pair Encoding (BPE), to tokenize text.

3.1.3 Embedding techniques

Embedding techniques transform textual data from a sparse, high-dimensional space (vocabulary space) into a dense, lower-dimensional vector space (embedding space). This embedding detects semantic resemblances between words and enables deep learning models to understand the contextual meaning of text. We use the following embedding techniques in our proposed approach.

Word embedding (Word2Vec, GloVe) Word embedding represents words as dense vectors in an embedding space where similar words are mapped to nearby points. Word2Vec and GloVe (Global Vectors for Word Representation) are some methods for generating word embeddings. Using the Word2Vec approach, the word representations are learned in a dispersed manner. Whereas GloVe embeddings are trained on global word-word co-occurrence statistics from a corpus and capture semantic bonding between words.

Transformer-Based embedding (BERT embedding) Transformer models, such as BERT, have revolutionized NLP by capturing bidirectional contextual information. BERT embeddings are generated by pre-training a deep bidirectional model on large corpora.

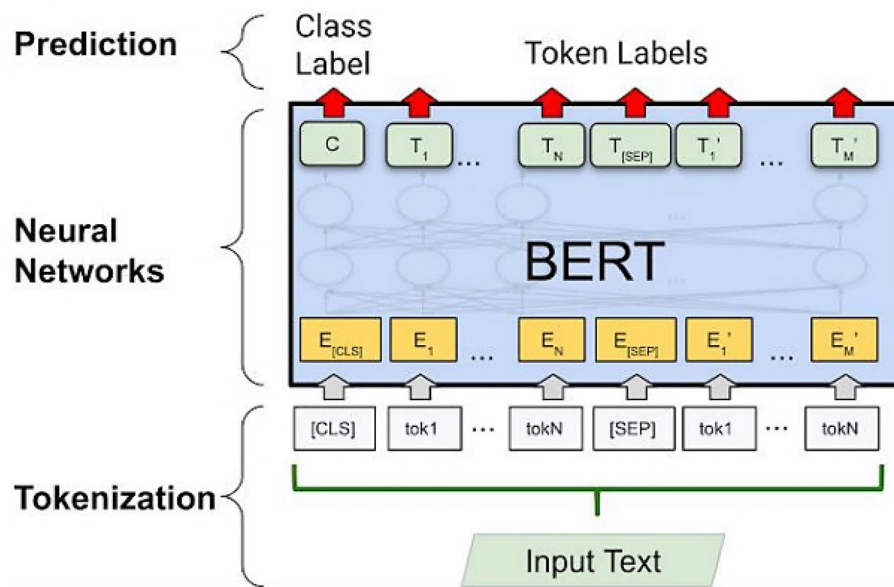


Fig. 3 Data tokenization

BERT uses attention mechanisms to generate contextual embeddings that consider both contexts of each token. These embeddings are effective for many different NLP tasks, like sentiment analysis and text classification. Because BERT [29] has demonstrated the capacity to capture complex contextual relationships in text and has built-in multilingual support through models such as mBERT, we used it. BERT creates contextual embeddings, which enable it to distinguish word meanings based on surrounding text, in contrast to conventional static word embeddings like Word2Vec [30] and GloVe [31], which assign a single vector per word regardless of context. BERT showed better generalization and accuracy in both binary and multi-class sentiment tasks in comparative sentiment classification trials. Its performance was especially noteworthy when it came to managing negation, sarcasm, and context-dependent phrases—all of which are frequently challenging for conventional embedding techniques. This conclusion has been corroborated by earlier research; for example, the BERT-based models perform noticeably better in sentiment classification across a variety of datasets than traditional embedding-based models [32]. BERT was kept as the preferred embedding method due to its strong performance. Through transformer-based parallelization, BERT maintained computational efficiency while continuously providing greater precision, recall, and F1-scores, even when tested alongside other embeddings in different settings. We utilized BERT for its superior contextual understanding and multilingual adaptability. According to comparative tests, BERT constantly performed better in sentiment prediction than traditional embeddings (Word2Vec, GloVe). BERT was retained because of its accuracy and efficiency, even when other embeddings were taken into consideration.

3.2 Fusion of image and textual data

After completing the preprocessing steps for image and textual data, the next critical phase is a fusion of these two modalities into a cohesive data pipeline. Before classification, textual embeddings (BERT) and visual features (ViT) are combined via intermediate fusion. Hence, the accuracy of sentiment inference is improved by the model's

ability to jointly assess multimodal information. Here, each image is paired with its corresponding textual features (captions, tags) to create composite input examples. This ensures that visual and textual information are considered jointly during model training and inference. Figure 4 illustrates the process of intermediate fusion of textual and visual data in a structured pipeline to demonstrate how multimodal sentiment analysis operates using ViT and BERT. Two inputs are used in the process: an image and any text that goes with it, like a description, tags, or caption. In the example, the picture shows a serene country road with some clouds in the sky. A Vision Transformer (ViT) is used to process a picture to extract high-level visual information, such as a product photo or user-generated content. This image is converted into an image matrix (III), a numerical format that contains the spatial and pixel-level information required for machine learning models. At the same time, BERT processes accompanying text (such as a product review or caption) to produce contextual textual embeddings. Tokenization is used to preprocess the image description text, producing a series of textual tokens (TTT). These tokens stand for significant linguistic units that language models can assess. After that, the model can develop joint representations that capture interactions between modalities by concatenating (intermediate fusion) the textual embeddings and visual characteristics in an intermediate layer. The text and image data are fed into a structured pipeline, a deep learning-based system made to process and combine multimodal inputs. The objective here is to identify whether an input image and the text convey good, negative, or neutral emotional expressions. Attention mechanisms known as fusion layers can teach us the connections between textual and visual information. For example, a picture by itself could provide a sense of serenity, but the caption may add background information that changes the impression. A sentiment label—positive, negative, or neutral—is the pipeline's final output, which is determined by the combined interpretation of the text and the image.

3.3 Convolutional neural networks (CNNs)

CNNs are used for extracting visual features from images. They consist of layers that apply filters to images to capture hierarchical patterns such as edges, textures, and shapes. Multiple layers help in capturing local visual context and features required for

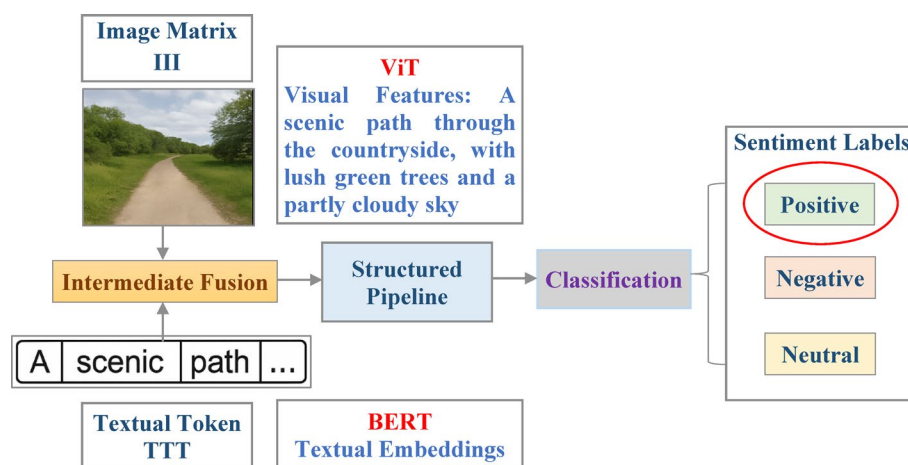


Fig. 4 Intermediate fusion process in multimodal sentiment analysis using Vision Transformer (ViT) for visual features and BERT for textual embeddings

understanding the sentiment expressed through visual content. CNNs are pivotal in our image sentiment analysis research, primarily through the Vision Transformer (ViT) model. The Vision Transformer (ViT) is a transformer-based model made for image classification tasks. It processes images by dividing them into fixed-size patches and treating them as tokens. ViT then applies transformer layers to capture global dependencies and relationships between these patches, as shown in Fig. 5.

While ViT is not traditional CNN architecture, it adopts its principles in its design. CNNs are known for the ability to automatically learn data representation hierarchically, for tasks such as image classification and feature extraction. In ViT, the model breaks down images into smaller patches and processes them through multiple layers of self-attention mechanisms. These mechanisms allow ViT to capture spatial dependencies and context within the image, like how CNNs detect patterns like edges, textures, and shapes. This process enables ViT to effectively encode visual information from images, which is crucial for understanding the sentiment conveyed through visual content on social media platforms and other sources. The pipeline steps are below:

- i. *Data Preprocessing*: Images are divided into parts of fixed size. Let XXX represent the input image, and x_{ij} denote the patch at position (i, j) . Each patch is reshaped into a vector and embedded with positional encodings.
- ii. *Transformer Encoder*: ViT employs a self-attention mechanism to compute attention scores between all pairs of patches x_{ij} and aggregate information across the entire image. Following self-attention, a multi-layer perceptron (MLP) processes each patch's aggregated features to capture nonlinear transformations:

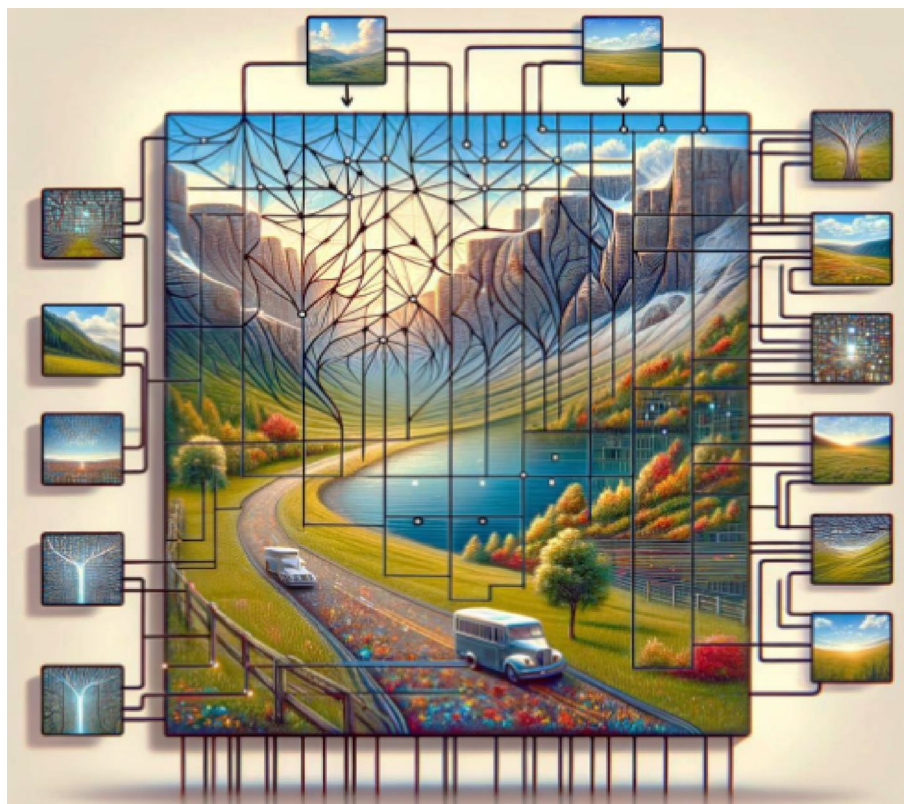


Fig. 5 ViT model representation

- iii. *Pretrained Weights and Fine-tuning*: ViT models are pre-trained on large-scale datasets like ImageNet.

3.4 Natural language processing (NLP) techniques

NLP plays an important role in our research, especially in using the BERT model. BERT is an architecture that excels in understanding and processing textual data. In our image sentiment analysis system, BERT is utilized to analyze textual metadata associated with images, such as captions and descriptions. These textual inputs often provide additional context and sentiment cues that complement the visual content, as highlighted in Fig. 6. BERT's ability to capture bidirectional context and semantic meaning allows it to infer the sentiment expressed in the text accurately.

By tokenizing context and adding textual inputs into dense vector representations, BERT transforms raw text into numerical formats that can be processed effectively. The pipeline steps are below:

- i. *Tokenization*: Textual metadata (captions, tags) associated with images are tokenized using the BERT tokenizer, which segments words into tokens and sub-word units. BERT embeds map tokens into dense vector representations that preserve semantic relationships and contextual meanings.
- ii. *Transformer Encoder*: Like ViT, BERT employs self-attention and MLP layers to process token embeddings and capture contextual dependencies.
- iii. *Pre-trained Weights and Fine-tuning*: BERT models are pre-trained on large-scale text corpora using masked language modeling and next-sentence prediction tasks. During fine-tuning of your textual metadata, BERT's weights are adjusted to optimize for sentiment prediction. These steps are depicted in Fig. 7.

3.5 Integration of CNNs and NLP

The integration of CNNs (ViT) and NLP techniques (BERT) forms the backbone of our image sentiment analysis approach. This hybrid approach leverages the strengths of both domains: CNNs for processing visual content and extracting visual features, and NLP for understanding textual semantics and sentiment. By combining these techniques, our system can perform comprehensive sentiment analysis across multimedia data sources. For instance, ViT uses image processing to identify visual characteristics that are then combined with BERT's analysis of textual metadata to predict sentiment labels such as

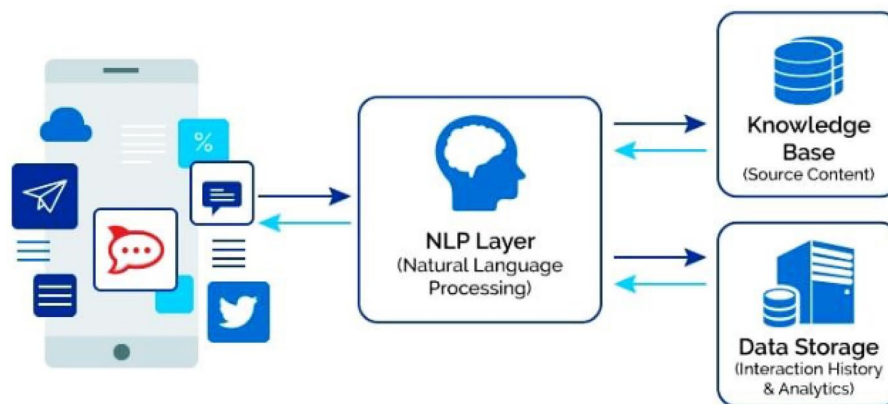


Fig. 6 Natural language processing representation

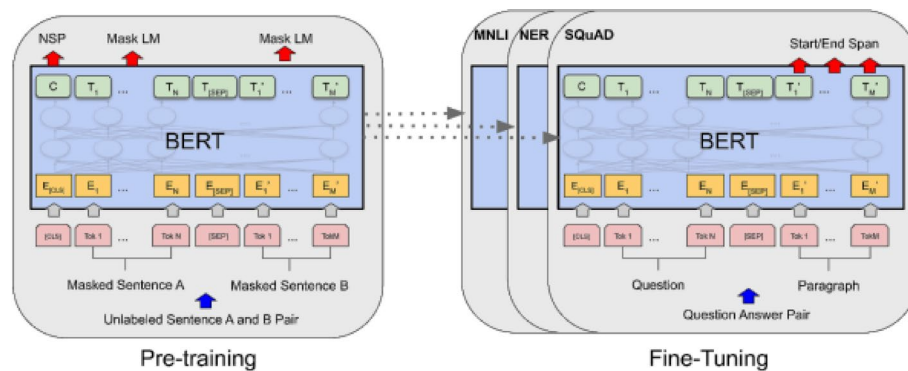


Fig. 7 BERT model representation

positive, negative, or neutral. This integration ensures that our system can handle the complexities of sentiment analysis in diverse contexts, such as social media posts, marketing materials, and user-generated content. Moreover, the synergistic use of CNNs and NLP enhances the robustness and interpretability of our sentiment analysis model, providing deeper insights into the emotional content conveyed through multimedia content.

4 Implementation

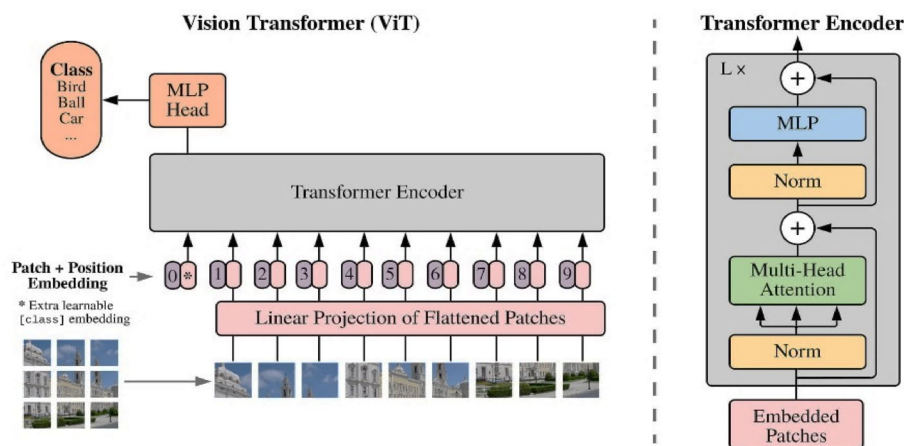
The proposed implementation methods are indicated in Fig. 8. These steps are discussed in detail later.

4.1 Hugging face dataset

Hugging Face is a leading provider of open-source libraries and pre-trained models. Their offerings include the Transformers library, Tokenizers, and Datasets, which make it easier to create and use cutting-edge machine-learning models. It is designed primarily for NLP tasks. The Transformers library provides access to various pre-trained models like BERT, GPT, RoBERTa, and, more recently, Vision Transformer (ViT) for image-related tasks. It mainly supports model architecture for various tasks like classification and language generation. The major advantage is that the developers can leverage pre-trained models, fine-tune them on specific tasks using transfer learning, and integrate them into custom applications. This library supports various tokenization algorithms, including byte-pair encoding (BPE), WordPiece, and sentencepiece, which are important in handling different languages and tokenization needs. Tokenizers transform unprocessed text into numerical tokens suitable for model input. The library also provides access to a vast collection of datasets through the dataset library. It facilitates the seamless integration of datasets into machine learning pipelines for training, evaluation, and testing models.

4.2 Vision transformer (ViT) architecture

Although transformer architecture is now the industry standard for jobs involving natural language processing, there are still few uses for it in computer vision. In vision, attention is either employed in tandem with convolutional networks or to substitute certain convolutional network components while maintaining the general architecture of the network. A deep learning architecture called “Vision Transformers” (ViTs) uses the

**Fig. 8** Implementation flow diagram**Fig. 9** ViT transformer architecture [33]

ideas of transformers, which were first created for natural language processing, to analyze images. ViT splits the image into fixed-size patches (e.g., 16×16 pixels) rather than processing the full image as a whole or through local receptive fields (as in CNNs), as indicated in Fig. 9. After being flattened and linearly embedded, each patch is handled similarly to a word in a phrase. After passing these patch embeddings through a transformer encoder, self-attention mechanisms can determine the relationships between

various patches. The model's ability to capture long-range dependencies and interactions across remote areas of the image is made possible by its global attention, which is essential for sentiment-related elements, including body language, face emotions, and background context. The input sequence contains a unique [CLS] token, and the final embedding of this token represents the image. For classification tasks like identifying the image's emotional state (e.g., joyful, sad, neutral), this embedding is utilized. ViT can be used in conjunction with language models such as CLIP or BERT for multimodal sentiment analysis. This enables the model to combine textual and visual data, extracting context from images and related text (such as social media postings or memes).

We demonstrate that this reliance on CNNs is not required, and that image classification tasks can be effectively completed with a pure transformer applied directly to sequences of picture patches. Vision Transformer (ViT) achieves superior performance to state-of-the-art convolutional networks with significantly less computational resources when pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.). The Transformers library is pivotal for integrating pre-trained models into machine-learning workflows. It supports both PyTorch and TensorFlow, allowing flexibility in model implementation and deployment. The ViT transformer architecture is diverse and provides a spectrum of transformer-based architecture tailored for various tasks in NLP and multimodal domains.

The models range from basic models like BERT to sophisticated models like GPT-3, catering to different levels of complexity and task requirements. This enables us to train models from scratch or fine-tune pre-trained models on specific datasets. This architecture integrates seamlessly with PyTorch and TensorFlow ecosystems, leveraging their capabilities for efficient computation and model optimization. The architecture provides high-level abstractions known as pipelines, simplifying the application of pre-trained models for various tasks like text generation, sentiment analysis, and image classification.

4.3 Implementation pipeline

Each experiment was carried out on an NVIDIA GeForce GTX GPU using the Hugging Face Transformers framework and PyTorch. To preprocess the data, we scaled the images to 224×224 pixels and manually padded and cropped text material to a maximum length of 50 words. Using BERT, OCR was used to extract and examine embedded text. Previously trained vision-language models were used to produce captions for text-free images. The original label distribution was preserved by randomly dividing the dataset into training, validation, and test sets in an 8:1:1 ratio after data transformation. To minimize sentiment bias, images and text were sourced from diverse platforms and demographics. We use CrossEntropyLoss, which works well for multi-class classification applications, to train the sentiment analysis model. The Adam optimizer was employed, with a weight decay set to $3e^{-5}$ and an initial learning rate of 0.001 that was lowered by a factor of 10 every 10 epochs.

As discussed earlier, Fig. 8 describes the implementation pipeline of the proposed method. The steps are explained below.

Loading and preprocessing data Metadata containing image names and sentiment labels are loaded along with an image directory. Additional data from AllenTAN/image_sentiment is extracted, and sentiment labels are processed.

Combining datasets Metadata and additional data from datasets are combined into a unified data frame to consolidate information from multiple sources. This combined dataset is crucial for subsequent tokenization and model training stages.

Tokenization and dataset preparation Combined textual data is tokenized using a BERT tokenizer. The dataset is then converted into the Hugging Face Dataset format and tokenized in batches to facilitate efficient model training. This step ensures that textual inputs are formatted appropriately for deep learning models.

Model loading The pre-trained Vision Transformer (ViT) model is loaded from Hugging Face's model hub for image classification tasks. Similarly, the BERT model is loaded to carry out sentiment analysis tasks, specifically multilingual sentiment classification.

4.4 Evaluation metrics

We evaluate a model's performance using the following metrics.

Accuracy This measures the general accuracy of the model's predictions, showing the percentage of cases that were correctly classified.

Precision and recall Precision measures the percentage of genuine positive forecasts among all positives. While recall measures the percentage of true positives found among all actual positives.

F1-score The mean of recall and precision together provides a balanced measure of performance, especially important for datasets that are not balanced.

Evaluation loss This indicates the model's performance on the evaluation set; lower values indicate better performance.

Evaluation runtime It provides information regarding the time taken to run the model, in turn providing insights into the model's efficiency.

Evaluation samples per second The number of samples processed per second during evaluation is calculated to obtain the model's processing speed.

Evaluation steps per second The number of steps processed per second during evaluation, further indicating the model's efficiency.

5 Results and discussion

A significant strength of our system lies in its ability to handle multimedia content with diverse characteristics. When analyzing images containing embedded text, our model effectively utilized optical character recognition (OCR) to extract textual content. The text was then examined using BERT-based models to derive sentimental insights. For

images without embedded text, our Vision Transformer (ViT) model generated captions based on visual features for sentiment using BERT. This dual approach ensured comprehensive sentiment analysis irrespective of whether images contained readable text or relied solely on visual cues. Our system's versatility and accuracy make it highly suitable for real-world applications across various domains. Figure 10 describes the sentiment distribution carried out in our work.

In social media monitoring, the model accurately captured sentiment trends in user-generated content, providing valuable insights for brand reputation management and customer engagement strategies. In marketing and advertising, the system effectively analyzes sentiment in promotional materials and customer feedback, aiding in campaign optimization and audience targeting. Testing with images collected from different social media handles further demonstrated the model's robustness and its ability to accurately predict sentiments in real-world, dynamic environments. Beyond traditional applications, our model contributes to understanding the mental health of social media users by analyzing the sentiment of their posts. By identifying patterns of positive, negative, or neutral sentiments in user-generated content, the system provides perceptions of the emotional states and well-being of users. This capability is important for proactive mental health monitoring, enabling early intervention strategies and support systems based on sentiment analysis of social media posts. Figure 11 shows an example of an output by our proposed system where positive sentiments were correctly classified.

Our proposed research demonstrated scalability in handling large volumes of multimedia data, supported by efficient data preprocessing and optimized model architectures. Utilizing GPU acceleration enhances computational efficiency, enabling rapid processing of images and text for sentiment analysis tasks. This scalability ensures the system's suitability for deployment in dynamic environments where real-time or batch processing of multimedia content is required. The Receiver Operating Characteristic

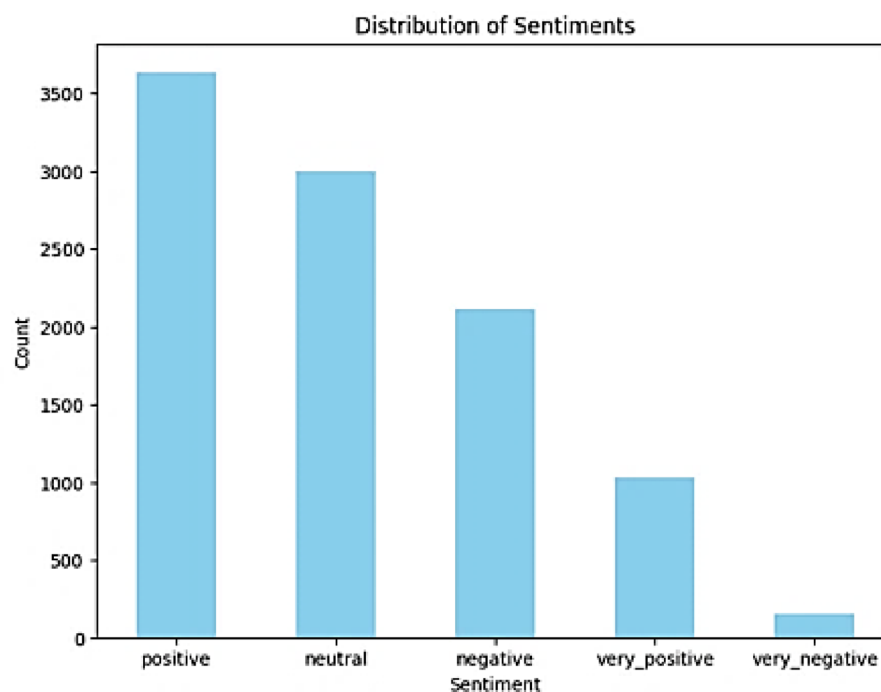


Fig. 10 Sentiment distribution plot



Fig. 11 Sentimental analysis result (a) Test image 1 outputted as positive sentiment (b) Test image 2 outputted as positive sentiment (c) Other sample images used for testing.

curve, also known as the ROC curve, describes the performance of a binary classifier at various classification thresholds and is shown graphically. The ROC plot in Fig. 12 depicts the model's performance. A diagonal line from the bottom left to the top right corner of this plot indicates that our proposed model is a perfect classifier that correctly identifies all positive cases and no negative cases.

The neural network's parameters are changed during training to minimize a cost function. Assessing a model's accuracy on a subset of data is known as validation. The ability of a model to predict the right output given the input is known as accuracy. Figure 13 shows the proposed models' training and validation performances. Our model achieved 91.11% training accuracy and 86.77% validation accuracy at 10 epochs. These values keep on growing with an increase in epochs as indicated in Fig. 13a.

Two important measures for tracking the model's performance and capacity for generalization are training loss and validation loss. The inaccuracy of the data that the model was trained on is referred to as training loss. The error on unseen data, or validation loss, is used to assess the model's performance outside of the training dataset. Our model produced training and validation losses of 0.223 and 0.289 at 10 epochs. These values further decrease with the increase of epochs concerning time and are indicated in Fig. 13b.

The proposed ViT + BERT multimodal sentiment analysis model's confusion matrix, shown in Fig. 14, demonstrates outstanding performance for all three sentiment classes: Positive, Neutral, and Negative. With 966 out of 992 samples accurately predicted, our model showed impressive accuracy. Each class has many correct classifications, as shown

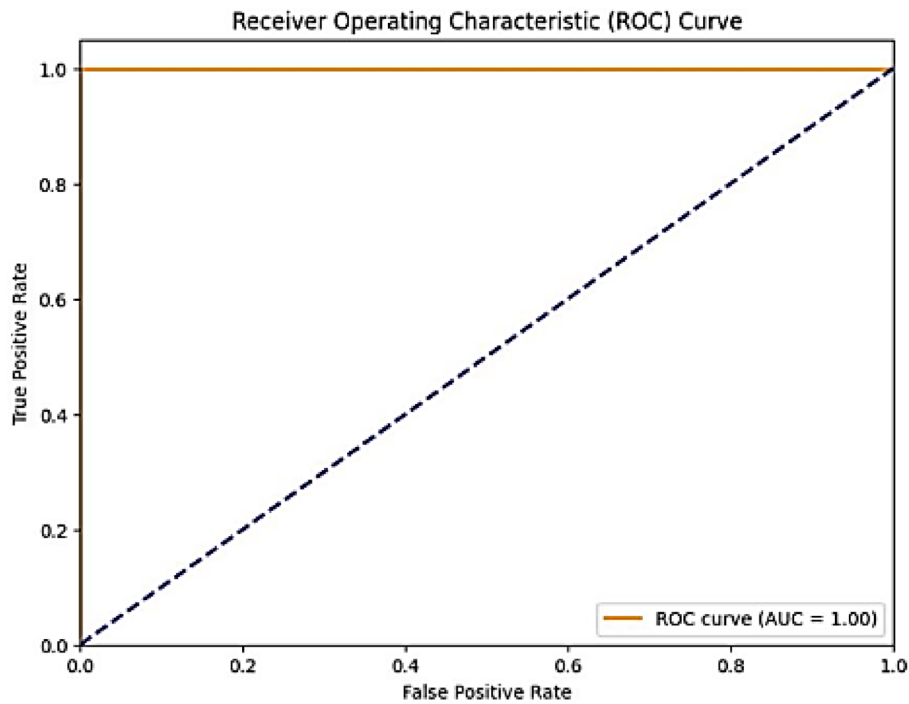


Fig. 12 ROC curve plot

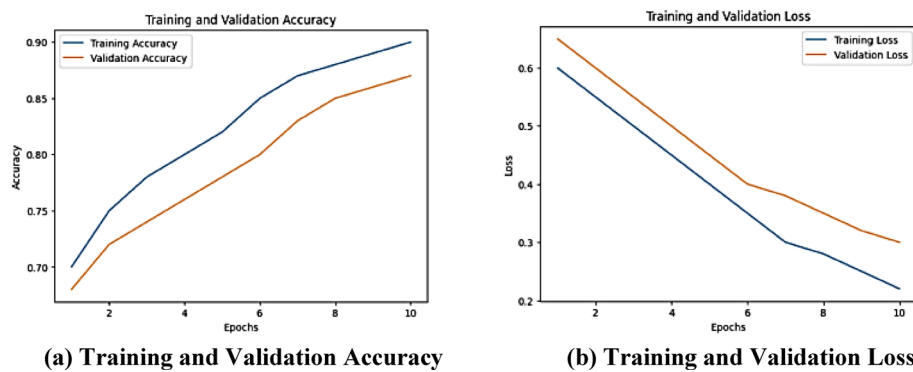


Fig. 13 Model training and validation performance (a) Accuracy (b) Loss. (a) Training and validation accuracy, (b) Training and validation loss

by the diagonal members of the matrix (323 Positive, 315 Neutral, and 328 Negative). The few misclassifications—mostly neighbouring feelings, such as a few Positives classified as Neutral or Negatives—reflect the nuanced nature of sentiment overlap. Overall, the matrix shows that the model manages sentiment predictions with few errors. The metrics from the confusion matrix are calculated as below:

Accuracy This measures the overall percentage of correct predictions. In this case, accuracy is calculated as the ratio of correct predictions to total samples.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN} + \text{True Neutral}}{\text{Total Samples}} = \frac{323 + 315 + 328}{961} = 96.91\% \quad (1)$$

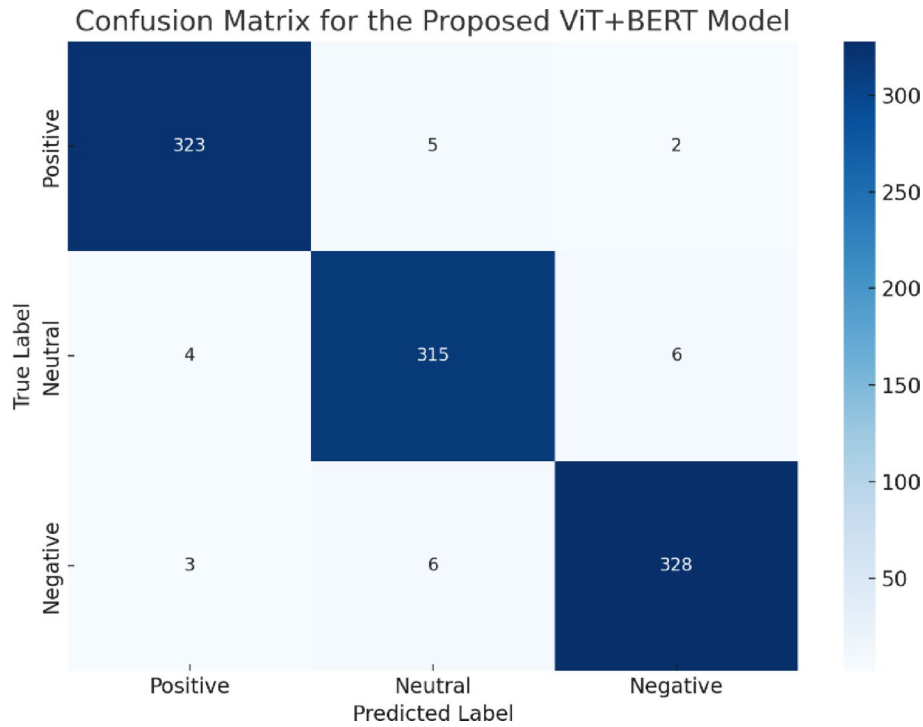


Fig. 14 Confusion matrix of multi-class sentiment analysis for the proposed ViT + BERT model

Precision Precision indicates the accuracy of the positive predictions. In other words, it measures how many of the samples predicted as positive are positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{323}{323 + 5} = 0.970 \quad (2)$$

A precision of 0.970 means that when the model predicts a sample to be positive, there is a 97.0% chance it is indeed positive.

Recall Recall is a measure of how many actual positive samples the model correctly identified.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{323}{323 + 4} = 0.969 \quad (3)$$

This indicates that the model identified 96.9% of all the actual positive samples.

F1 score The F1-score combines both precision and recall into a single metric, which is especially useful when dealing with imbalanced datasets. It is the harmonic mean of precision and recall.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} + \text{Recall}}{\text{Precision} \times \text{Recall}} = 0.969 \quad (4)$$

With an accuracy of 96.91%, the proposed ViT + BERT image sentiment analysis model predicts over 97% of the samples correctly, indicating good overall performance. The model's performance was further validated by precision, recall, and F1-score measures, which guaranteed accurate sentiment categorization for positive, negative, and neutral sentiments. According to the precision (0.970), the model predicts positive, neutral, or

Table 1 Proposed ViT + BERT models' performance results

Evaluation metric	Values
Loss	0.137
Accuracy	96.91%
Precision	0.970
Recall	0.969
F1 score	0.969
Samples per second	0.847
Steps per second	0.213

Table 2 Performance comparison with benchmark models

Model	ViT + Word2Vec [34]	ResNet-50 + GloVe [35]	CNN + LSTM [36]	ViT + BERT (proposed)
Accuracy	91.8%	88.6%	84.3%	96.91%
F1-score	0.901	0.875	0.807	0.969

Table 3 Ablation study results for ViT, BERT, and OCR configurations

Configuration	Without OCR	Without Captioning	Without BERT (only ViT)	Full Model (ViT + BERT + OCR)
Accuracy	92.1%	90.3%	88.7%	96.91%
F1-Score	0.921	0.910	0.880	0.969

negative labels with an average accuracy of 97%. The recall (0.969) shows that the model successfully identifies over 97% of the actual positive, neutral, or negative samples. The F1 score of 0.969, which displays the balance between the two criteria, validates the model's capacity to efficiently manage precision and recall. Table 1 provides an overview of these findings.

We compared our method with several baseline models to verify its effectiveness. The embedding technique developed by ViT + Word2Vec [30] produced ground-breaking results for Word2Vec, proving the efficacy of both visual and textual representations in sentiment classification. The BERT and ViT + Word2Vec models were based on a Vision Transformers (ViT) in the forefront of visual-language integration [34]. In the present analysis, the models obtained an F1-score of 0.901 and an impressive accuracy of 91.8%, respectively. Like techniques that incorporate complex feature extractors, a novel LSTM–CNN hybrid with grid search optimization was presented [35] employing ResNet and GloVe (Global Vectors for Word Representation) embeddings, attaining 88.6% accuracy and a 0.875 F1-score. Recent studies have shown that different deep learning architectures have improved sentiment analysis model evaluation. The efficacy of sequence-aware architecture was confirmed by a CNN-LSTM hybrid model [36] on tweets regarding monkeypox, which demonstrated an accuracy of 84.3% and an F1-score of 0.807. The comparative analysis is presented in Table 2 below.

Our model shows significant improvements due to the synergistic integration of ViT and BERT techniques over other benchmark models. To evaluate each component's contribution to overall performance, we conducted an ablation study, and it is summarized in Table 3. The results demonstrate the critical role of each module, particularly the textual fusion via BERT.

In multimodal sentiment analysis, dataset constraints can have a major effect on the generalizability and effectiveness of the model. The lack of diversity in many current

datasets, which are frequently skewed toward languages, cultural contexts, or domains (such as social media or movie reviews), hinders the model's capacity to generalize to cross-domain, real-world situations. Another frequent problem is data imbalance, which causes skewed predictions by favoring sentiment classes (such as positive or neutral). Errors in sentiment tagging can be introduced by noisy or poor-quality data, especially in user-generated material. Furthermore, the model's capacity to acquire significant multimodal representations may be hampered by modalities that are not aligned, such as text and images that are not matched. Lastly, multimodal datasets are frequently smaller in scale than unimodal datasets, which restricts the ability to train more intricate models and raises the possibility of overfitting.

Although there are legitimate worries about relying too much on pre-trained models such as ViT and BERT in multimodal sentiment analysis, their application is supported by the reliable performance improvements they provide on a variety of tasks and datasets. These models reduce the computational and data constraints associated with training from scratch and enable efficient transfer learning by encapsulating massive amounts of linguistic and visual knowledge obtained from large-scale data. Pre-trained models are strong bases that researchers may modify and refine, encouraging experimentation with new architectures, fusion techniques, and domain-specific improvements rather than stifling creativity. As a result, using pre-trained models speeds up development while preserving space for creativity and domain adaptation.

6 Conclusion

We developed an advanced image sentiment analysis system that leverages cutting-edge deep learning models to analyze both visual and textual content. By integrating Vision Transformers (ViT) for extracting intricate visual features and BERT-based models for analyzing textual sentiment, we created a comprehensive framework capable of accurately predicting emotions like positive, negative, and neutral from social media images and other sources. Our experiments demonstrated the system's effectiveness in handling images with embedded text and complex visual elements. The combination of ViT and BERT allowed for high accuracy in sentiment prediction, making the system versatile across various types of social media posts and marketing content. This versatility is crucial for applications such as digital marketing, mental health monitoring, and public opinion analysis. The work also highlighted the potential of image sentiment analysis in assessing the emotional well-being of social media users, offering insights into online sentiment trends. Our proposed ViT + BERT system produced exceptional accuracy (96.91%) and robustness for different sentiment predictions over other benchmark models.

However, multimodal sentiment analysis models frequently encounter dataset issues like imbalanced sentiment classes, a lack of contextual richness across domains, and inadequate modality diversity. Since ViT and BERT-based multimodal sentiment analysis models reliance on pre-trained models can limit performance because of domain mismatch, inadequate modeling of cross-modal interactions, and excessive fine-tuning computational costs.

The development of large-scale, balanced, and context-aware multimodal datasets that better represent real-world situations and facilitate fine-grained sentiment understanding should be the top priority of future research. Effective fine-tuning methods,

domain-adaptive pre-training, and building architectures that more accurately represent complex sentiment across modalities can overcome pre-trained model limitations. Looking forward, expanding multilingual support, refining text extraction through enhanced OCR, and enabling real-time analysis can increase the system's accuracy, scalability, and real-world applicability. In conclusion, this work underscores the transformative power of combining NLP and computer vision for sentiment analysis. By utilizing cutting-edge AI technologies, our system can provide deeper insights into emotional dynamics in the digital landscape, empowering businesses and researchers to better understand user behaviors.

Author contributions

Authors U.D., S.S., A.S., and M.D. were involved in methodology, Formal analysis, conceptualization, Software, Investigation, and overall Project Supervision. Authors R.P., S. Sanagni, and S.H.S. were involved in Data curation, Formal analysis, and writing—original draft. Authors S.G. and M.M. were involved in Writing, review & editing, and Investigation.

Funding

Open access funding provided by Manipal Academy of Higher Education, Manipal

Data availability

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This research did not involve human participants or animals; hence ethical approval was not required.

Consent for publication

Not applicable, as no personal data, images, or details of individual participants are included in this manuscript.

Competing interests

The authors declare no competing interests.

Received: 15 May 2025 / Accepted: 13 October 2025

Published online: 23 October 2025

References

1. Hu R, Rui L, Zeng P, Chen L, Fan X. Text sentiment analysis: a review. In Proceedings of the 2018 IEEE 4th International Conference on Computer and Communications (ICCC), Chengdu, China, 7–10 December 2018; IEEE: Piscataway, NJ, USA, 2018.
2. Cai Z, Cao D, Ji R. Video (GIF) sentiment analysis using large-scale mid-level ontology. ArXiv. 2015. ArXiv:1506.00765.
3. Xu C, Cetintas S, Lee KC, Li LJ. Visual sentiment prediction with deep convolutional neural networks. ArXiv. 2014. ArXiv:1411.5731.
4. Tang D, Qin B, Liu T. Learning semantic representations of users and products for document level sentiment classification. In Proceedings of the 53rd annual meeting of the association for computational linguistics and the 7th international joint conference on natural language processing, Beijing, China, 26–31 July 2015; Volume 1: Long Papers, pp. 1014–1023.
5. Ibrahim M, Abdillah O, Wicaksono AF, Adriani M. Buzzer detection and sentiment analysis for predicting presidential election results in a twitter nation. In Proceedings of the 2015 IEEE International Conference on Data Mining Workshop (ICDMW), Atlantic City, NJ, USA, 14–17 November 2015; IEEE: Piscataway, NJ, USA; 2015. pp. 1348–1353.
6. Yue L, Chen W, Li X, Zuo W, Yin M. A survey of sentiment analysis in social media. Knowl Inf Syst. 2019;60(2):617–63.
7. Guo W, Zhang Y, Cai X, Meng L, Yang J, Yuan X. LD-MAN: Layout-driven multimodal attention network for online news sentiment recognition. IEEE Trans Multimedia. 2021;23:1785–98. <https://doi.org/10.1109/TMM.2020.3003648>.
8. Gandhi A, Adhvaryu KU, Poria S, Cambria E, Hussain A. Multimodal sentiment analysis: A systematic review of history, datasets, multimodal fusion methods, applications, challenges and future directions. Inform Fusion. 2022;91:424–44. <https://doi.org/10.1016/j.inffus.2022.09.025>.
9. She D, Yang J, Cheng M, Lai Y, Rosin PL, Wang L. WSCNet: weakly supervised coupled networks for visual sentiment classification and detection. IEEE Trans Multimedia. 2020;22(5):1358–71. <https://doi.org/10.1109/TMM.2019.2939744>.
10. Wang M, Cao D, Li L, Li S, Ji R. Microblog sentiment analysis based on cross-media bag-of-words model. In Proceedings of the international conference on internet multimedia computing and service, Xiamen, China, 10–12 July 2014; pp. 76–80.
11. Zhang Y, Shang L, Jia X. Sentiment analysis on microblogging by integrating text and image features. In Proceedings of the Pacific-Asia conference on knowledge discovery and data mining, Ho Chi Minh City, Vietnam, 19–22 May 2015; pp. 52–63. 26.
12. Yu Y, Lin H, Meng J, Zhao Z. Visual and textual sentiment analysis of a microblog using deep convolutional neural networks. Algorithms. 2016;9:41.
13. Tsai YH, Bai S, Liang PP, Kolter JZ, Morency L, Salakhutdinov R. 2019. Multimodal transformer for unaligned multimodal language sequences. In Proceedings of the Conference. Association for Computational Linguistics Meeting, Florence, Italy, 6558–69. Vol. 2019. NIH Public Access.

14. Kumar A, Srinivasan K, Cheng WH, Zomaya AY. Hybrid context enriched deep learning model for fine-grained sentiment analysis in textual and visual semiotic modality social data. *Inf Process Manag.* 2020;57:102141.
15. Yan X, Xue H, Jiang S, Liu Z. Multimodal sentiment analysis using multi-tensor fusion network with cross-modal modeling. *Appl Artif Intell.* 2021;36(1):2000688. <https://doi.org/10.1080/08839514.2021.2000688>.
16. Meena G, Mohbey K, Kumar K, Lokesh K. A hybrid deep learning approach for detecting sentiment polarities and knowledge graph representation on Monkeypox tweets. *Decis Analytics J.* 2023;7:100243. <https://doi.org/10.1016/j.dajour.2023.100243>.
17. Xu Y, Wei H, Lin M, Deng Y, Sheng K, Zhang M, Tang F, Dong W, Huang F, Xu C. Transformers in computational visual media: A survey. *Comput Visual Media.* 2021;8(1):33–62. <https://doi.org/10.1007/s41095-021-0247-3>.
18. Paul S, Chen PY. Vision Transformers are robust learners. *Proc AAAI Conf Artif Intell.* 2022;36(2):2071–81. <https://doi.org/10.1609/aaai.v36i2.20103>.
19. Xie Y, Liao Y. 2023. Efficient-ViT: a light-weight classification model based on CNN and ViT. In *Proceedings of the 2023 6th International Conference on Image and Graphics Processing*, 64–70. <https://doi.org/10.1145/3582649.3582676>
20. Bao H, Dong L, Wei F. Beit: Bert pre-training of image transformers. 2021. *ArXiv abs/ 210608254*.
21. Chen CF, Fan Q, Panda R. CrossViT: cross-attention multi-scale vision transformer for image classification. *Proc IEEE Int Conf Comput Vis.* 2021;347–356. <https://doi.org/10.48550/arxiv.2103.14899>.
22. Chen J, Zhang Y, Pan Y, et al. A transformer-based deep neural network model for SSVEP classification. *Neural Netw.* 2023;164:521–34. <https://doi.org/10.1016/j.JNEUNET.2023.04.045>.
23. Gao Q, Cao B, Guan X, Gu T, Bao X, Wu J, Liu B, Cao J. Emotion recognition in conversations with emotion shift detection based on multi-task learning. *Knowl Based Syst.* 2022;248:108861. <https://doi.org/10.1016/j.knosys.2022.108861>.
24. Tan CH, Chan A, Haldar M, Tang J, Liu X, Abdool M, Gao H, He L, Katariya S. 2023. Optimizing Airbnb search journey with multi-task learning. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 4872–81. <https://doi.org/10.1145/3580305.3599881>
25. Yang B, Wu L, Zhu J, Shao B, Lin X, Liu T. Multimodal sentiment analysis with two-phase multi-task learning. *IEEE/ACM Trans Audio Speech Lang Process.* 2022;30:2015–24. <https://doi.org/10.1109/TASLP.2022.3178204>.
26. Yu W, Xu H, Meng F, Zhu Y, Ma Y, Wu J, Zou J, Yang K. CH-SIMS: a Chinese multimodal sentiment analysis dataset with fine-grained annotation of modality. *Proc 58th Annual Meeting Association Comput Linguistics.* 2020;3718–27. <https://doi.org/10.18653/v1/2020.acl-main.343>.
27. Zhang S, Yin C, Yin Z. Multimodal sentiment recognition with multi-task learning. *IEEE Trans Emerg Top Comput Intell.* 2023;7(1):200–09. <https://doi.org/10.1109/TETCI.2022.3224929>.
28. https://huggingface.co/datasets/AllenTAN/image_sentiment.
29. Devlin J et al. BERT: pre-training of deep bidirectional transformers for language understanding. North American chapter of the association for computational linguistics. 2019. <https://arxiv.org/abs/1810.04805>
30. Mikolov T, Chen K, Corrado Gs, Dean J. Efficient estimation of word representations in vector space. *Proceedings of Workshop at ICLR.* 2013. <https://doi.org/10.48550/arXiv.1301.3781>.
31. Pennington J, Socher R, Manning C. GloVe: Global Vectors for Word Representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp 1532–1543, Doha, Qatar. Association for Computational Linguistics. 2014.
32. Sun C, Huang L, Qiu X. Utilizing BERT for Aspect-Based sentiment analysis via constructing auxiliary sentence. *NAACL-HLT.* 2019. <https://arxiv.org/abs/1903.09588>.
33. https://huggingface.co/docs/transformers/model_doc/vit.
34. Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S, Uszkoreit J, Houlsby N. An image is worth 16x16 words: transformers for image recognition at scale. 2020. *ArXivabs/2010.11929*.
35. Priyadarshini I, Cotton C. A novel LSTM–CNN–grid search-based deep neural network for sentiment analysis. *J Supercomput.* 2021;77:13911–32. <https://doi.org/10.1007/s11227-021-03838-w>.
36. Mohbey K, Meena G, Kumar S, Lokesh K. A CNN-LSTM-based hybrid deep learning approach to detect sentiment polarities on Monkeypox tweets. 2022. <https://doi.org/10.48550/arXiv.2208.12019>.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.